



Course name	Network Architectures and Protocols / Computer networks
The topic of the document	Laboratory exercise book for university students
Laboratory exercise book no.	4.
The title and the topic of the laboratory exercise	Addressing in IPv4 part 1. Mapping of logical and physical address

1. Preparation and implementation requirements:

- The student is obliged to read in advance the content of the present laboratory exercise book, to do the preparations for the laboratory and to observe the rules defined for the laboratory.
- The students should read the description of the special terms, mechanisms listed in the „Theoretical background information” chapter to accomplish the laboratory tasks successfully.
- The appearance happens at the beginning of the laboratory, delay can be allowed only in justified cases.
- The use of students’ own mobile phone and PC is forbidden in the laboratory, except the case when the laboratory leader permits their use for the students because of the type of the laboratory exercise to be done.
- Browsers can be used exclusively for the access to information which is necessary for the accomplishment of the laboratory exercise, in the case of the laboratory leader’s permission or instruction.
- By the end of the class the laboratory leader evaluates each participant’s activity based on the composed questions which are included in this present material and further possible questions.
- The laboratory will be successful, if the student provides a correct answer to each question on the basis of the laboratory exercises which are carried out on the premises.
- The replacement of the laboratory is possible at dates of time determined by the laboratory leader.

2. Theoretical background information:

During the laboratory the following special terms will be used in practice:

- Broadcast address – ppt slides of the lecture, slide no. 24.
- Broadcast domain – ppt slides of the lecture, slide no. 24.
- Data link layer, network layer – ppt slides of the lecture, slide no. 39.
- Physical address – ppt slides of the lecture, slide no. 102,
https://en.wikipedia.org/wiki/MAC_address
- Addressing in IP – ppt slides of the lecture no. 142-238.
- Virtualization, virtual machines –
<https://www.virtualbox.org/manual/UserManual.html>
- Linux commands:



- arp – <https://man7.org/linux/man-pages/man8/arp.8.html>
- ping – <https://man7.org/linux/man-pages/man8/ping.8.html>
- ifconfig – <https://man7.org/linux/man-pages/man8/ifconfig.8.html>
- Windows commands
 - arp – <https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/arp>
 - ping – <https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/ping>
 - ipconfig – <https://docs.microsoft.com/en-us/windows-server/administration/windows-commands/ipconfig>

During the laboratory the following mechanisms, special techniques will be used:

- Creating a new virtual machine in VirtualBox – https://docs.oracle.com/cd/E26217_01/E26796/html/qs-create-vm.html
- Creating a new virtual network and configuring a VM – <https://www.techrepublic.com/article/how-to-create-multiple-nat-networks-in-virtualbox/>
- IPv4 masking – ppt slides of the lecture, slide no. 177-181.
- Physical address is determined from network address (ARP) and vica versa (RARP) – ppt slides of the lecture no. 183-186.
- Physical address and OUI – <https://aruljohn.com/mac.pl>

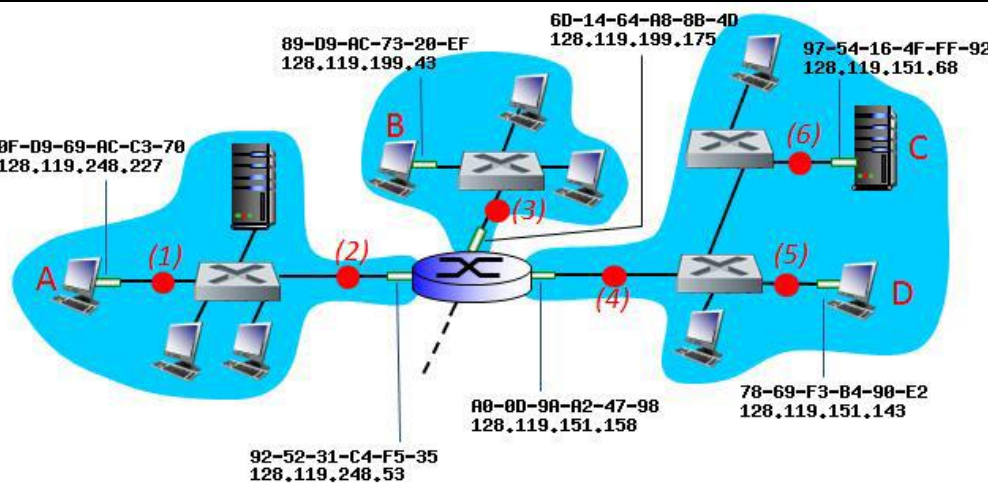
3. The following tasks have to be performed on the site:

3.1.	Masking of IP address, calculation of network address/netid and host address/hostid
	• Generate a random IP address with the help of the following page: https://www.tunnelsup.com/subnet-calculator/ Avoid using netmasks /8, /16 and /24 (in slash notation). • With the help of the generated IP address and netmask determine the related network address/netid and host address/hostid. • Determine the class of the obtained IP address.

3.2.	IP address configuration of a VM
	• If you did not do it earlier, create a new virtual machine in VirtualBox. Run Knoppix operational system on the PC. • Create a new virtual private network with netid/prefix length 10.0.0.0/8 and connect the virtual machine to this network. • Create two additional virtual machines (or create a clone about the former one) and configure their network connections to the same network.

3.3.	Physical address and ARP table
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	• Display the physical address of the first virtual machine. • With the help of https://aruljohn.com/mac.pl webpage search the manufacturing company name which is related to the OUI code of the physical address. • Display the arp table of the 1st VM. • „Ping” the 1st VM from the 2nd VM and the 3rd one, then display again the arp table. • Analyse the change of the arp tables. • Shutdown the network connection of the 1st VM, then reconnect it. • Analyse the change of the arp table. • Insert a new entry to the table (arp -s). • Delete a table entry (arp -d). • Repeat the operations on the host as well (under Windows) that can be executed on one VM.
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3.4.	We send a datagram from node B to node A. What will be the source and destination host addresses (MAC address and IP address) in the Ethernet frame at points (3), (2) and (1) according to the figure? (the boxes represent the switch, the cylinder represent the router)
	

4. The following questions have to be answered by the end of the laboratory:

4.1.	A) Network address/Netid:
	B) Host address/Hostid:
	C) IP address class:
	D) Netmask related to the address / (in slash notation) and in dotted decimal form:



4.2.	A) What IP addresses did the virtual machines receive in the network?
	B) How would be possible to change the IP address of the 1st machine for the IP address used in the previous exercise?

4.3.	A) What is the physical address of the 1st machine? How can you display this information (by using only linux terminal)?
	B) What is the name of the manufacturing company assigned to this physical address?
	C) What does happen with the ARP table after shutting down the connection and after reconnection?
	D) Which command did you use precisely to insert a new entry to the arp table?
	E) Which command did you use precisely to delete a static entry from the arp table?

4.4.	A) Fill in the following table:		
		Source	Destination
	(3)	IP	
		MAC	
	(2)	IP	
		MAC	
	(1)	IP	
		MAC	
	B) Explain what is the role of the physical address and that of the IP address in the network communication. Which type of address can be used in what cases (when)?		